

A VISUAL INTRODUCTION

A Field Guide to Finite Element Families

Why there is more than one kind of finite element,
and how to tell them apart

	space	element	what is glued	typical PDE
	H^1	Lagrange P_k	the whole value	Poisson, elasticity, heat
	$H(\text{curl})$	Nédélec N_k	tangential part $v \cdot t$	Maxwell, eddy currents
	$H(\text{div})$	RT_k , BDM_k	normal part $v \cdot n$	Darcy, mixed Poisson
	L^2	discontinuous P_k	nothing	advection, shocks

reading down the table: less and less is glued together across element boundaries

Companion notes to the falk-mixed-elasticity repository.
Everything in this document is reproducible: see README.md.

Before you start

This guide is written for someone who is comfortable with calculus and linear algebra — gradients, divergence, integration by parts, vector spaces, bases — but who has never sat down and studied finite element spaces. You do not need to have met Sobolev spaces, weak formulations or the *inf-sup* condition before; each of those is introduced here when it is first needed, in plain language before any notation appears.

The question the guide answers is this one:

Software like FEniCSx offers you Lagrange, DG, RT, BDM, N1curl and a dozen more. Why are there so many? What actually distinguishes them, and how do you know which one your equation wants?

The short answer, which the rest of the guide unpacks, is that the families differ in **how much of a function is forced to match across the boundary between two neighbouring cells** — and that the right amount of matching is dictated by the partial differential equation you are solving, not by taste.

One practical note on reading: almost every diagram in this guide shows the *same* two triangles meeting along one edge. That is deliberate. Once you have the picture of two cells and their shared edge in your head, every element family becomes a different answer to the single question “what happens on that edge?”.

A note on indices: this guide uses the classical mathematical convention in which RT_0 and N_0 denote the *lowest-order* Raviart–Thomas and Nédélec spaces. Some software, including Basix/FEniCSx, counts from one instead, so the lowest-order space is called degree 1 there. The elements are the same; only the label shifts by one.

Contents

Before you start	1
1 What is a finite element?	3
1.1 Ingredient one: the mesh	3
1.2 Ingredient two: a polynomial on each cell	3
1.3 Ingredient three: degrees of freedom	4
1.4 Assembling cells into a global space	4
2 Why there is more than one kind of element	5
2.1 Different equations, different unknowns	5
2.2 Weak formulations, and why they decide everything	5
2.3 The one calculation that explains all of it	6
2.4 The same calculation for vector fields	6
2.5 What this buys us	7
3 Lagrange elements: gluing the value	8
3.1 The element	8
3.2 Why point values at shared nodes give continuity	8
3.3 The model problem	8
3.4 The physical reading	9
4 Discontinuous Galerkin: gluing nothing	9
4.1 The element	9
4.2 If nothing is glued, what holds the solution together?	10

4.3	Why anyone would want this	11
4.4	The other role of DG: never being differentiated	11
5	Raviart–Thomas: gluing the normal component	11
5.1	The problem that demands it	11
5.2	The element	12
5.3	What normal continuity means physically	13
5.4	Why plain Lagrange elements are the wrong tool here	13
6	BDM: the same continuity, a different polynomial space	14
6.1	The element	14
6.2	The essential difference from RT	14
6.3	When to prefer which	14
7	Nédélec elements: gluing the tangential component	15
7.1	The problem that demands it	15
7.2	The element	15
7.3	What tangential continuity means physically	15
7.4	Why Lagrange elements fail here — and they really do fail	15
8	Four more elements worth knowing	16
8.1	Crouzeix–Raviart: continuity at one point per edge	16
8.2	Hermite and Argyris: when H^1 is not enough	17
8.3	Taylor–Hood: elements come in pairs	17
9	One table for all of them	19
10	The de Rham complex: why this is a family, not a list	19
10.1	The sequence	19
10.2	The discrete copy	20
10.3	What exactness buys	20
10.4	Two complexes in two dimensions, and where BDM sits	21
11	Putting it together	21
11.1	A worked example: one element, three spaces	21
11.2	How to choose, in practice	22
11.3	The one-page summary	22
	Where to go next	23

1 What is a finite element?

Suppose you want to solve a partial differential equation on some region Ω — the temperature in a turbine blade, the pressure in an aquifer, the electric field in a waveguide. The unknown is a function, and a function carries infinitely much information. A computer needs finitely many numbers. The finite element method is one recipe for choosing which finitely many numbers to keep.

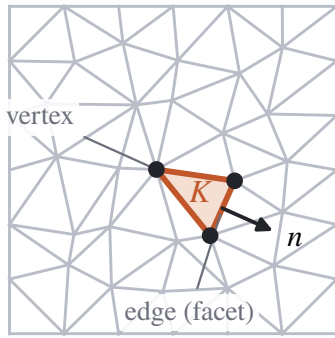
The recipe has three ingredients, and it is worth separating them carefully, because almost all the variety between element families lives in the third one.

1.1 Ingredient one: the mesh

Chop Ω into small, simple pieces: triangles or quadrilaterals in two dimensions, tetrahedra or hexahedra in three. Call the collection $\mathcal{T}_h$, an individual piece a *cell* or *element*, and write h for the typical cell diameter. The pieces must fit together exactly — no gaps, no overlaps, and (for the standard theory) a vertex of one triangle may not land in the middle of another triangle's edge.

Two cells that share an edge are *neighbours*, and the edge they share is called a *facet* (an edge in 2D, a triangular face in 3D). Facets are where all the interesting behaviour happens.

(a) a mesh of the domain Ω



(b) the patch used in every later figure

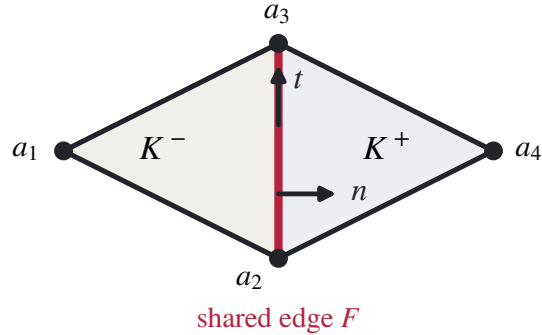


Figure 1. Left: a mesh of a square domain, with one cell K highlighted and the outward unit normal $\mathbf{n}$ on one of its edges. Right: the two-cell patch that every later figure in this guide reuses. The shared edge F carries a normal direction $\mathbf{n}$ (pointing across it) and a tangential direction $\mathbf{t}$ (pointing along it).

1.2 Ingredient two: a polynomial on each cell

On each individual cell we approximate the unknown by a polynomial of low degree. Two reasons make polynomials the natural choice. Practically, they are trivial to differentiate and to integrate exactly by quadrature. Theoretically, Taylor's theorem says that any smooth function looks like a polynomial of degree k when you zoom in far enough, with an error of size $O(h^{k+1})$ on a cell of diameter h . So refining the mesh and raising the degree are two different knobs for the same goal.

Write $P_k(K)$ for the polynomials of total degree at most k on the cell K . In two dimensions

$$P_1 = \text{span}\{1, x, y\}, \quad P_2 = \text{span}\{1, x, y, x^2, xy, y^2\},$$

of dimension 3 and 6; in general $\dim P_k = \frac{1}{2}(k+1)(k+2)$ in 2D.

1.3 Ingredient three: degrees of freedom

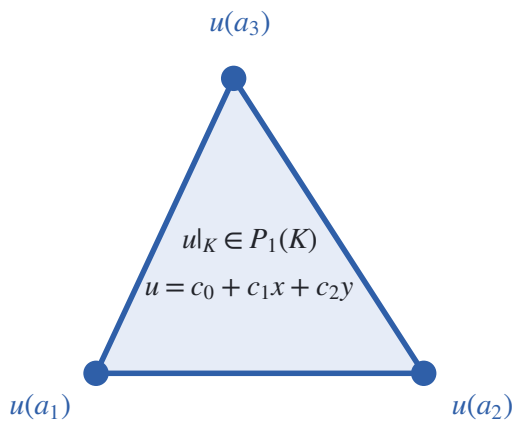
Here is where the families separate. A polynomial on a cell is determined by its coefficients, but coefficients are a terrible bookkeeping device: they have no geometric meaning, and two neighbouring cells have no way to agree on them. Instead we pin the polynomial down by *degrees of freedom*.

Degree of freedom

A *degree of freedom* (dof) is a rule that extracts one number from a function. “Its value at this vertex.” “Its average over this edge.” “The flux of this vector field through this edge.” Formally it is a linear functional. A set of dofs is good if it is *unisolvent*: knowing all the numbers determines the polynomial uniquely.

For P_1 on a triangle, the classical choice is the value at each of the three vertices. Three numbers, three dimensions, and a linear function in the plane is indeed fixed by its values at three non-collinear points — unisolvent.

(a) three values fix a linear function



(b) the hat function φ_i

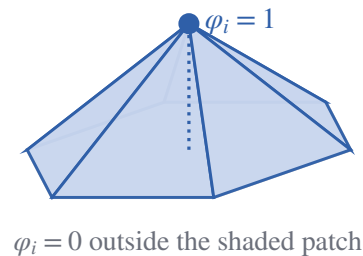


Figure 2. Left: the three degrees of freedom of the P_1 Lagrange triangle. Right: the corresponding global basis function, the “hat” φ_i , which equals 1 at one vertex, 0 at every other vertex, and vanishes outside the cells touching that vertex. Small support is what makes the resulting matrices sparse.

The formal definition, due to Ciarlet, packages exactly these three ingredients: a finite element is a triple (K, P, Σ) — a cell, a space of polynomials on it, and a unisolvent set of dofs. When people say “the Raviart–Thomas element” they mean a particular choice of P and Σ .

1.4 Assembling cells into a global space

Now the step that matters most. We have a polynomial on every cell; we want one function on all of Ω . The rule is simple and it is the key to this whole guide:

Degrees of freedom that two neighbouring cells share are stored once. A dof sitting on a vertex or an edge belongs to *every* cell touching that vertex or edge, so all of them are forced to agree on it. A dof sitting strictly inside a cell belongs to that cell alone, and no neighbour ever sees it.

That single bookkeeping decision — where you attach the dofs — determines what is continuous across facets, and therefore which global function space you end up in. Put the dofs on vertices and you glue values together. Put them on edges and you glue something edge-related together. Put them strictly inside and you glue nothing.

The rest of this guide is, quite literally, a catalogue of the different answers to “what do we glue?”.

Finally, a bookkeeping payoff: because each dof is shared by only a handful of cells, the corresponding basis function (Figure 2, right) is nonzero only on those few cells. Two basis functions whose supports do not overlap contribute a zero to the stiffness matrix, so the matrix is sparse — which is the reason the method scales to millions of unknowns.

2 Why there is more than one kind of element

If gluing more is always safer, why not always glue everything? Because the equation you are solving decides how much gluing is *necessary*, and gluing more than necessary is at best wasteful and at worst simply wrong.

2.1 Different equations, different unknowns

The unknowns in real problems are not all scalar fields:

Scalar fields. Temperature, pressure, concentration, electrostatic potential. One number per point.

Vector fields. Displacement of an elastic body, fluid velocity, electric and magnetic fields. Several numbers per point — and, crucially, the components may not all behave the same way.

Fluxes. The Darcy velocity in a porous rock, the heat flux in a composite. These are vector fields whose entire physical meaning is “how much stuff crosses this surface per unit time”.

Tensor fields. The stress in a solid. In the companion repository to this guide, the stress is the primary unknown.

Quantities that genuinely jump. The density across a shock wave, the concentration across a contact discontinuity. Forcing these to be continuous is not conservative caution — it is a modelling error.

2.2 Weak formulations, and why they decide everything

Finite elements do not solve the PDE pointwise. They solve its *weak* (or variational) form.

Weak formulation

Multiply the equation by an arbitrary smooth *test function* v , integrate over Ω , and use integration by parts to move derivatives off the unknown and onto v . The result is a family of integral identities, one per test function. A function satisfying all of them is a *weak solution*. The point is that integrals tolerate functions that are far less smooth than the original PDE seems to require — kinks, and sometimes jumps.

Take the Poisson equation $-\Delta u = f$. Multiply by v vanishing on $\partial\Omega$ and integrate by parts once:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx.$$

Look at what survives: exactly one derivative on u , and it appears inside a square-integrable pairing. So the natural home for u is the set of functions whose *first derivatives are square-integrable*. That set has a name.

The spaces L^2 , H^1 , $H(\text{div})$, $H(\text{curl})$

$L^2(\Omega)$ is the set of functions with $\int_{\Omega} u^2 < \infty$ — finite energy, no smoothness required at all. The

others add exactly one derivative, but only the derivative combination the PDE actually uses:

$$\begin{aligned} H^1 &= \{u : u \in L^2, \nabla u \in L^2\} && \text{all first derivatives} \\ H(\text{div}) &= \{\mathbf{v} : \mathbf{v} \in L^2, \text{div } \mathbf{v} \in L^2\} && \text{only the divergence} \\ H(\text{curl}) &= \{\mathbf{v} : \mathbf{v} \in L^2, \text{curl } \mathbf{v} \in L^2\} && \text{only the curl} \end{aligned}$$

$H(\text{div})$ and $H(\text{curl})$ are strictly *larger* than $(H^1)^d$: they ask for less, because the PDE asks for less.

Conformity

A discrete space V_h is *conforming* for a problem posed in V if $V_h \subset V$. Conformity is what lets you reuse the whole theory of the continuous problem. The entire design effort behind *RT*, *BDM* and *Nédélec* elements goes into making piecewise polynomials land inside $H(\text{div})$ or $H(\text{curl})$, which — as we are about to see — is a statement about facets.

2.3 The one calculation that explains all of it

Here is the computation that generates the whole catalogue. Take a function u_h that is a nice polynomial inside each cell but may jump across facets, and ask: what is its derivative, understood in the weak sense?

Integrate by parts cell by cell against a smooth test function φ that vanishes near $\partial\Omega$:

$$\int_{\Omega} u_h \partial_i \varphi \, dx = \sum_{K \in \mathcal{T}_h} \left(- \int_K \partial_i u_h \varphi \, dx + \int_{\partial K} u_h \varphi n_i \, ds \right).$$

Every interior facet F is visited twice, once from each side, with opposite normals. Writing u^- and u^+ for the two one-sided limits and $[[u]] := u^- - u^+$ for the *jump*, the boundary terms collapse to

$$\int_{\Omega} u_h \partial_i \varphi \, dx = - \underbrace{\sum_K \int_K \partial_i u_h \varphi \, dx}_{\text{the piecewise derivative}} + \underbrace{\sum_F \int_F [[u_h]] \varphi n_i \, ds}_{\text{a term living on the facets}}. \quad (1)$$

The first term is an ordinary L^2 function. The second is not a function at all: it is a measure concentrated on a surface, a Dirac delta smeared over the facet. Its “value” is infinite on a set of zero volume.

So, reading (1) backwards:

A piecewise polynomial lies in H^1 **if and only if** $[[u_h]] = 0$ on every interior facet — that is, if and only if it is continuous.

2.4 The same calculation for vector fields

Now repeat it for the divergence. With $\mathbf{v}_h$ piecewise polynomial and φ scalar,

$$\int_{\Omega} \mathbf{v}_h \cdot \nabla \varphi \, dx = - \sum_K \int_K (\text{div } \mathbf{v}_h) \varphi \, dx + \sum_F \int_F [[\mathbf{v}_h \cdot \mathbf{n}]] \varphi \, ds.$$

Only the combination $\mathbf{v}_h \cdot \mathbf{n}$ appears. The tangential part of $\mathbf{v}_h$ never enters. Hence:

A piecewise polynomial vector field lies in $H(\text{div})$ **if and only if** its *normal* component $\mathbf{v} \cdot \mathbf{n}$ is continuous across every facet. The tangential component is free to jump.

The curl version works the same way, using $\nabla \cdot (\mathbf{v} \times \varphi) = (\text{curl } \mathbf{v}) \cdot \varphi - \mathbf{v} \cdot (\text{curl } \varphi)$, and produces the mirror-image conclusion:

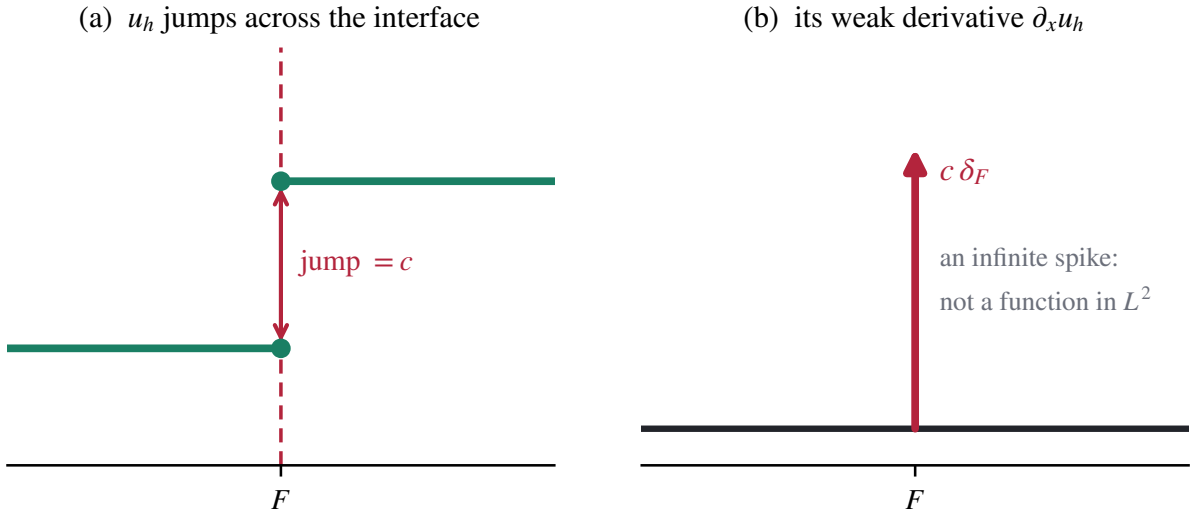


Figure 3. Why a jump throws you out of H^1 . A piecewise constant function with a jump of size c across the facet F has weak derivative equal to zero away from F plus an infinite spike of strength c on F . A spike is not a square-integrable function, so the jumping function is not in H^1 — no matter how smooth it is inside each cell.

A piecewise polynomial vector field lies in $H(\text{curl})$ **if and only if** its *tangential* component is continuous across every facet. The normal component is free to jump.

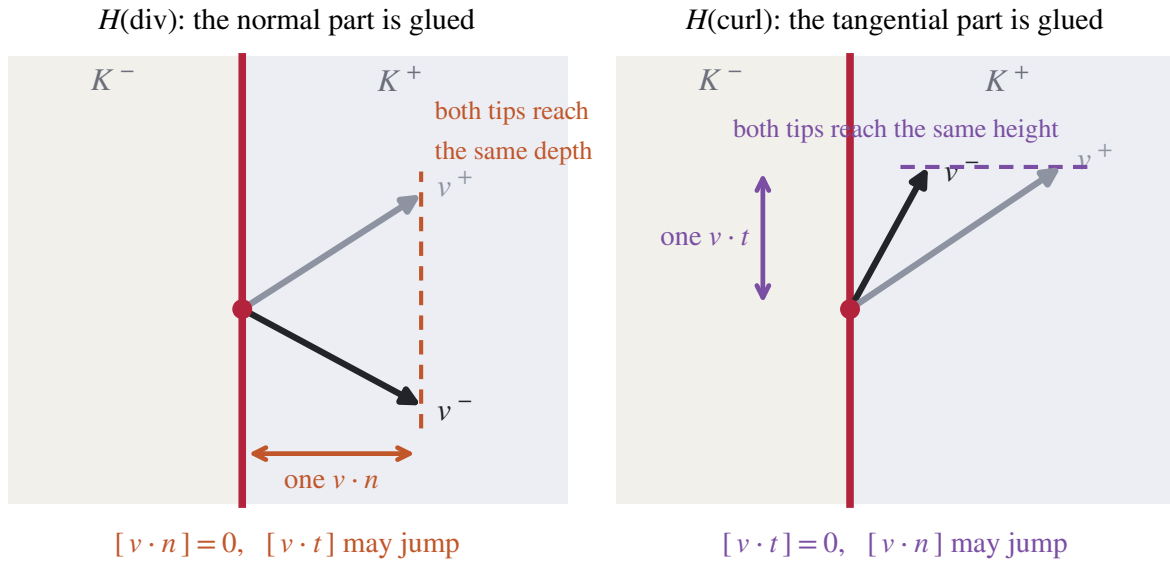


Figure 4. The central distinction of this guide. At a point on the shared edge, a vector field has two one-sided limits, v^- from the left cell and v^+ from the right. In an $H(\text{div})$ field (left) the two arrows reach equally far *across* the edge — the normal components agree — while their along-edge parts differ. In an $H(\text{curl})$ field (right) the two arrows reach equally far *along* the edge and differ across it. Neither field is continuous as a vector; each is continuous in exactly one component.

2.5 What this buys us

Three observations close the section.

First, H^1 -conformity is the *strongest* of the three conditions: it forces both components to match.

$H(\text{div})$ and $H(\text{curl})$ each force one. L^2 forces none. So $(H^1)^d \subset H(\text{div}) \cap H(\text{curl})$, and both of those sit inside L^2 .

Second, the conditions on $H(\text{div})$ and $H(\text{curl})$ are not comparable with each other. Neither is stronger. They are siblings.

Third — and this is the practical punchline — *demanding more continuity than the PDE needs is not a safe default*. It shrinks the space, so it costs accuracy per unknown; and when the true solution genuinely has a jump in the component you glued, it makes the method converge to the wrong answer. Sections 5 and 7 both contain concrete instances.

3 Lagrange elements: gluing the value

Approximates: a scalar field (or each component of a vector field independently). **Dofs:** point values at nodes. **Continuous:** the whole value. **Lives in:** H^1 . **Used for:** Poisson, heat, elasticity — anything whose weak form contains $\nabla u \cdot \nabla v$.

3.1 The element

The Lagrange element of degree k takes $P_k(K)$ on each cell and uses point values at a lattice of nodes as its dofs:

- P_1 : the 3 vertices.
- P_2 : the 3 vertices and the 3 edge midpoints — 6 nodes, and $\dim P_2 = 6$.
- P_3 : 3 vertices, 2 points on each of the 3 edges, 1 interior point — 10 nodes, and $\dim P_3 = 10$.

The pattern generalises: vertices first, then interior-of-edge nodes, then interior-of-face nodes, and so on.

3.2 Why point values at shared nodes give continuity

This is worth doing carefully once, because the argument recurs for every family in a different guise.

Take P_2 and two triangles sharing an edge F . Restricted to F , a function in P_2 of two variables becomes a polynomial of degree 2 in one variable — and a one-dimensional quadratic is determined by three values. The three nodes sitting on F (two endpoints plus the midpoint) supply exactly those three values, and both cells are forced to use the same numbers. Therefore the two traces on F coincide identically, not merely at the nodes.

That is the mechanism in general: the dofs lying on a facet must be enough to determine the *trace* on that facet, and they must be shared.

Note what the right-hand panel of Figure 5 also shows: the derivative *does* jump. That is fine. H^1 requires $\nabla u \in L^2$, and a function with a finite jump in its slope has a perfectly good bounded gradient — the gradient jumps, but it never becomes a spike. Only a jump in the *value* produces the Dirac term of Figure 3.

3.3 The model problem

For $-\Delta u = f$ with $u = 0$ on the boundary, set

$$V_h = \{ v_h \text{ continuous, piecewise } P_k, v_h|_{\partial\Omega} = 0 \},$$

and solve: find $u_h \in V_h$ with

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h \, dx = \int_{\Omega} f v_h \, dx \quad \text{for all } v_h \in V_h.$$

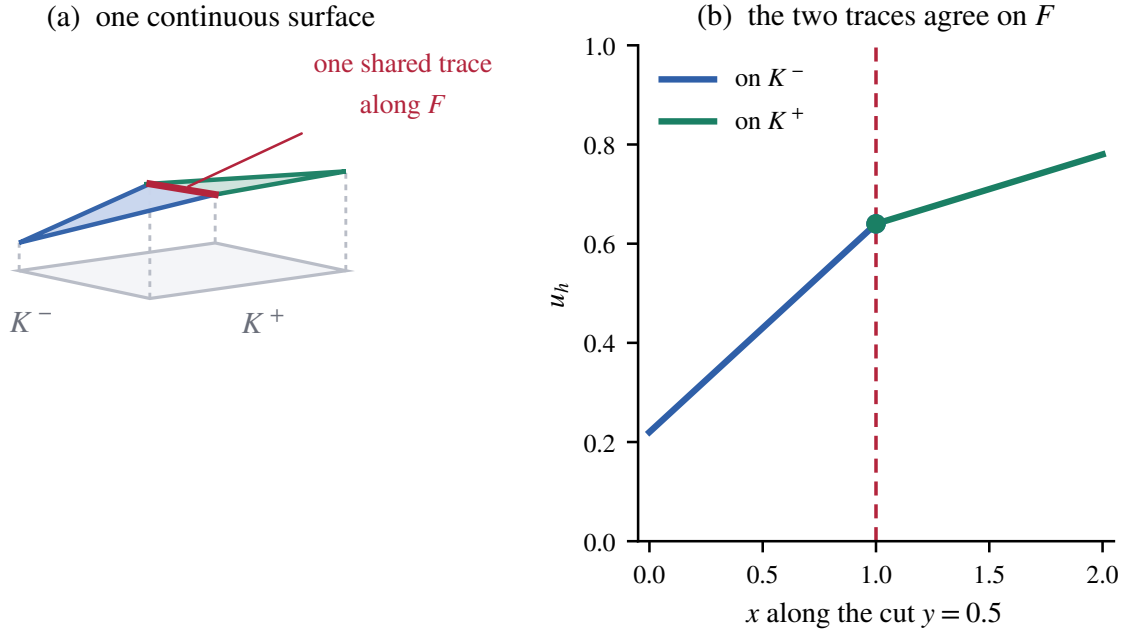


Figure 5. A P_1 Lagrange function on the two-cell patch. The two triangles share the values at both endpoints of the common edge, so the two linear traces along that edge are the same function: the graph is a single unbroken surface. Right: a cut through the patch. The function is continuous, but its slope is not — C^0 , not C^1 , which is exactly what H^1 asks for.

Because $V_h \subset H_0^1$, this is a conforming method, and the classical error estimates apply: for a smooth solution,

$$\|u - u_h\|_{H^1} = O(h^k), \quad \|u - u_h\|_{L^2} = O(h^{k+1}).$$

3.4 The physical reading

H^1 -continuity is the mathematical statement of “this quantity cannot jump”. Temperature cannot jump across an imaginary surface drawn inside a solid. Displacement cannot jump inside an intact material — a jump would mean the material has torn. When your unknown is a quantity of that kind, Lagrange elements are not merely convenient; they encode the physics.

The flip side is the subject of the next four sections: when the unknown is *not* of that kind, Lagrange elements impose a constraint the physics does not have.

4 Discontinuous Galerkin: gluing nothing

Approximates: a scalar or vector field that may genuinely jump, or one that is never differentiated. **Dofs:** cell-local, shared with nobody. **Continuous:** nothing. **Lives in:** L^2 . **Used for:** conservation laws and transport; also as the “other half” of mixed methods.

4.1 The element

Take the same P_k polynomials as before — and refuse to share anything. Each cell carries its own independent copy, so the global dimension is simply (number of cells) $\times$ $\dim P_k$. Nothing links a cell to its neighbour.

The dofs are usually still point values at nodes, but the nodes are understood as belonging to that cell alone. Two cells meeting at a vertex have two separate values there, and nothing forces them to agree.

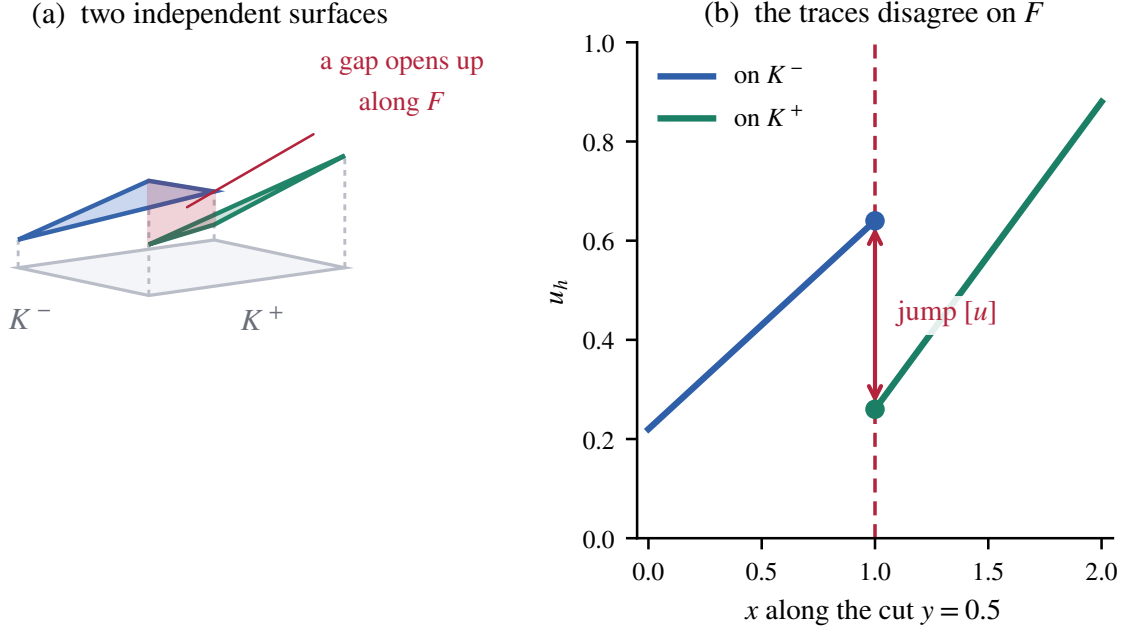


Figure 6. A discontinuous P_1 function on the same patch. The two cells now carry unrelated linear functions, so the two traces along the shared edge differ and the graph tears open. The size of the tear is the jump $[[u_h]] = u^- - u^+$, which for DG methods is not an error to be eliminated but a quantity the method actively works with.

4.2 If nothing is glued, what holds the solution together?

A good question, and the answer is the defining idea of the method: the cells are coupled *through the equations*, not through the space. Since the pieces do not agree at a facet, the method has to choose what value to use there — and that choice is called a *numerical flux*.

Jump, average, numerical flux

On a facet with two one-sided limits $u^\pm$, define $[[u]] = u^- - u^+$ and $\{\{u\}\} = \frac{1}{2}(u^- + u^+)$. A *numerical flux* $\hat{u}$ is a single value, built from u^- and u^+ , that both cells agree to use in place of the true trace. Different choices give different methods.

For the transport equation $\partial_t u + \nabla \cdot (\mathbf{b}u) = 0$, multiply by a test function supported on one cell and integrate by parts on that cell alone:

$$\frac{d}{dt} \int_K u_h v dx - \int_K u_h \mathbf{b} \cdot \nabla v dx + \int_{\partial K} \hat{u} (\mathbf{b} \cdot \mathbf{n}) v ds = 0.$$

The natural choice for $\hat{u}$ is *upwinding*: take the value from whichever side the flow is coming from,

$$\hat{u} = \begin{cases} u^- & \text{if } \mathbf{b} \cdot \mathbf{n} > 0, \\ u^+ & \text{if } \mathbf{b} \cdot \mathbf{n} < 0. \end{cases}$$

This is not a trick; it is the numerical statement that information in a transport problem travels downstream. Taking the average instead ($\hat{u} = \{\{u\}\}$) gives a scheme that is unstable, for the same reason that central differencing is unstable for advection.

For elliptic problems one wants the discontinuities to be small, so an *interior penalty* term

$$\sum_F \frac{\sigma}{h} \int_F [[u_h]] [[v_h]] ds$$

is added to the formulation. It pushes the jumps towards zero as the mesh refines without ever forbidding them outright: continuity is enforced *weakly* instead of being built into the space.

4.3 Why anyone would want this

- **Genuine discontinuities.** The solution of a nonlinear conservation law develops shocks in finite time even from smooth data. A continuous space cannot represent a shock and responds with spurious oscillations near it.
- **Local conservation.** Taking $v \equiv 1$ on a single cell turns the DG equation into an exact statement that what flows in minus what flows out equals the change inside *that cell*. Conservation holds cell by cell, not merely globally.
- **A block-diagonal mass matrix.** Because no basis function is shared between cells, the mass matrix is block diagonal with one small block per cell, and can be inverted cell by cell. For explicit time-stepping this is a decisive advantage.
- **Easy adaptivity.** Neighbouring cells need not use the same polynomial degree and meshes need not match across an interface, since nothing has to line up.

The cost is unknowns: a DG P_1 space on a triangular mesh has roughly six times as many dofs as continuous P_1 on the same mesh.

4.4 The other role of DG: never being differentiated

There is a second, quieter reason to use L^2 spaces. In the mixed methods of the next two sections, one of the unknowns is never differentiated in the weak form at all. If no derivative ever acts on it, then by the logic of Section 2 there is no reason to demand any continuity, and asking for some would only shrink the space for nothing. Such unknowns belong in a discontinuous space — and that is exactly where mixed methods put them.

5 Raviart–Thomas: gluing the normal component

Approximates: a flux — a vector field whose meaning is “how much crosses this surface”. **Dofs:** fluxes through facets. **Continuous:** the normal component $\mathbf{v} \cdot \mathbf{n}$. **Lives in:** $H(\text{div})$. **Used for:** Darcy flow, mixed Poisson, and every problem where local conservation matters.

5.1 The problem that demands it

Groundwater flow through a porous rock is governed by two statements. Darcy’s law says the flow follows the pressure gradient, damped by the permeability K of the rock; conservation of mass says nothing is created or destroyed:

$$\mathbf{u} = -K\nabla p, \quad \text{div } \mathbf{u} = f.$$

You could eliminate $\mathbf{u}$ and solve $-\text{div}(K\nabla p) = f$ with Lagrange elements. Often that is the wrong move, because in this application $\mathbf{u}$ — not p — is what you actually want, and differentiating a computed p_h to get it loses an order of accuracy and, worse, loses conservation.

So keep both unknowns. Multiply Darcy’s law by a vector test function $\mathbf{v}$, multiply the mass balance by a scalar test function q , and integrate the pressure term by parts:

$$\begin{aligned} \int_{\Omega} K^{-1} \mathbf{u} \cdot \mathbf{v} \, dx - \int_{\Omega} p \, \text{div } \mathbf{v} \, dx &= - \int_{\partial\Omega} p \mathbf{v} \cdot \mathbf{n} \, ds, \\ \int_{\Omega} q \, \text{div } \mathbf{u} \, dx &= \int_{\Omega} f q \, dx. \end{aligned} \tag{2}$$

Read off what the weak form demands. Derivatives hit $\mathbf{u}$ and $\mathbf{v}$ only through div — so $\mathbf{u} \in H(\text{div})$, and nothing more is required. And p is never differentiated at all — so $p \in L^2$, and asking for more would be gratuitous. The two spaces of a mixed method fall straight out of the integration by parts.

5.2 The element

The lowest-order Raviart–Thomas space on a triangle is

$$RT_0(K) = \left\{ \mathbf{v}(\mathbf{x}) = \mathbf{a} + b\mathbf{x} : \mathbf{a} \in \mathbb{R}^2, b \in \mathbb{R} \right\},$$

a three-dimensional space: two constants plus one radial term. The dofs are the three fluxes

$$\mathbf{v} \mapsto \int_{F_i} \mathbf{v} \cdot \mathbf{n}_i \, ds, \quad i = 1, 2, 3.$$

Why does this work? Here is the one-line miracle. Fix an edge lying on the line $\{\mathbf{x} : \mathbf{x} \cdot \mathbf{n} = c\}$. For $\mathbf{v} = \mathbf{a} + b\mathbf{x}$,

$$\mathbf{v} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} + b(\mathbf{x} \cdot \mathbf{n}) = \mathbf{a} \cdot \mathbf{n} + bc = \text{constant on that edge}.$$

The radial term $b\mathbf{x}$, which is not constant as a vector field, nevertheless has a *constant normal component along each straight edge*. So the normal trace on each edge is a single number, one dof pins it down, and the two cells sharing an edge are forced to agree on it. That is $H(\text{div})$ -conformity, built by hand.

Higher order follows the same idea:

$$RT_k(K) = (P_k)^2 \oplus \mathbf{x} \tilde{P}_k, \quad \dim RT_k = (k+1)(k+3),$$

where $\tilde{P}_k$ denotes the homogeneous polynomials of degree exactly k . The normal trace on each edge is a polynomial of degree k , fixed by $k+1$ dofs per edge; the remaining $k(k+1)$ dofs are interior moments.

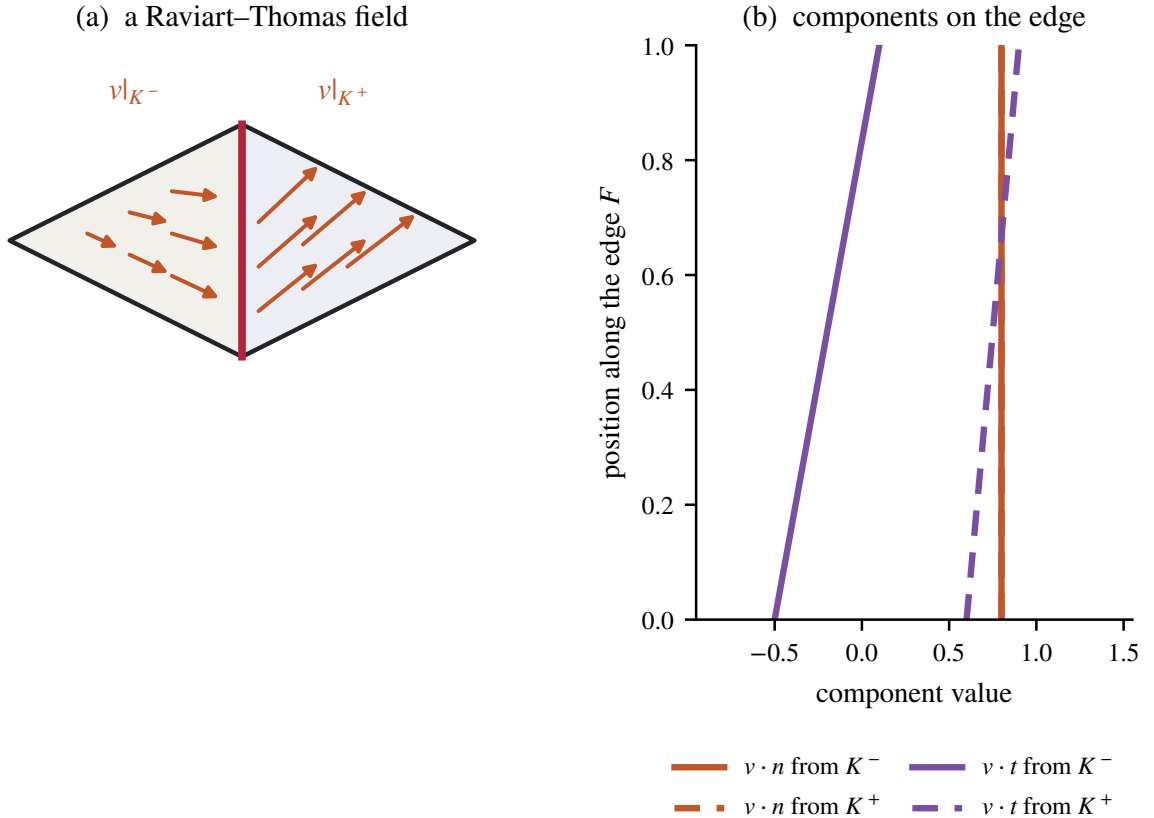


Figure 7. A Raviart–Thomas field on the two-cell patch. Left: the arrows visibly kink as they cross the shared edge — this is *not* a continuous vector field. Right: resolving the two one-sided traces into components along the edge shows what is really going on. The two normal-component curves (orange, solid from K^- and dashed from K^+) lie exactly on top of one another, while the two tangential curves (purple) are completely different functions.

5.3 What normal continuity means physically

Flux. The quantity $\int_F \mathbf{u} \cdot \mathbf{n}$ is the amount of fluid crossing the facet F per unit time. A jump in $\mathbf{u} \cdot \mathbf{n}$ would mean that the amount leaving one cell differs from the amount entering the next — mass appearing or vanishing on a surface of zero volume.

Normal continuity therefore *is* conservation, and in the discrete method it is exact rather than approximate. Take $q = 1$ on a single cell K and 0 elsewhere in the second equation of (2); the divergence theorem turns it into

$$\int_{\partial K} \mathbf{u}_h \cdot \mathbf{n} ds = \int_K f dx,$$

which holds to machine precision, on every cell, on every mesh, however coarse. Reservoir and transport codes care about this a great deal: an “accurate” flow field that leaks mass will corrupt the concentration field it is coupled to.

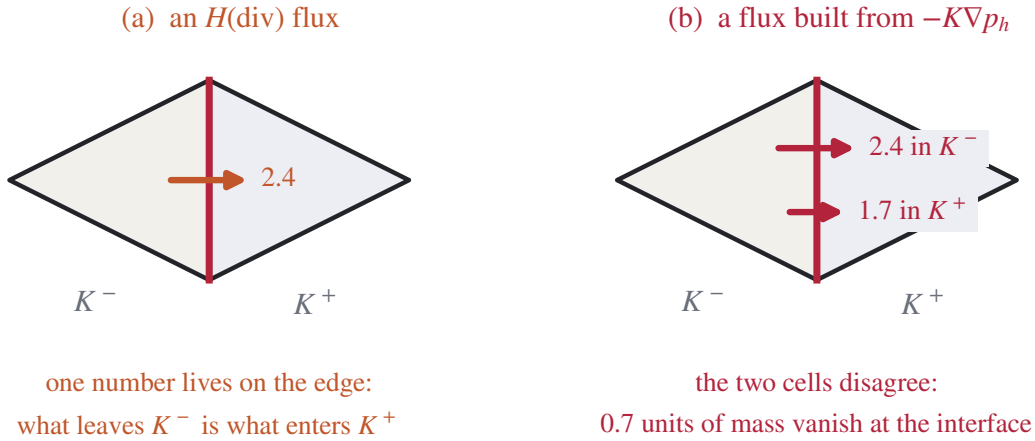


Figure 8. Local conservation. With an $H(\text{div})$ -conforming flux there is exactly one flux value on the shared edge and it is shared by both cells. A flux recovered by differentiating a continuous pressure, $-K\nabla p_h$, has two different values on that edge — and the difference is mass that the scheme creates or destroys at the interface.

5.4 Why plain Lagrange elements are the wrong tool here

Three independent reasons, each sufficient on its own.

1. **The physics has a tangential jump.** Suppose the permeability K jumps across a geological layer boundary, as it always does. The pressure p is continuous, so its tangential derivative along the interface is continuous, and therefore the tangential velocity $\mathbf{u} \cdot \mathbf{t} = -K \partial_t p$ jumps by the same ratio as K . Meanwhile $\mathbf{u} \cdot \mathbf{n}$ stays continuous, because mass is conserved. The true solution is thus normally continuous and tangentially discontinuous — precisely an $H(\text{div})$ field. A continuous vector Lagrange space forces *both* components to match and cannot represent it; it will smear the layer over one cell and converge slowly or to the wrong thing.
2. **No local conservation.** A flux post-processed as $-K\nabla p_h$ from a Lagrange pressure has two different values on each facet (Figure 8), so cellwise mass balance fails.
3. **Stability.** The mixed system (2) is a saddle point problem, and not every pair of spaces gives an invertible system. RT_k paired with discontinuous P_k is one of the classical stable pairs; obvious alternatives such as continuous $(P_1)^2$ with discontinuous P_0 are not. The inf-sup condition behind this is discussed in Section 8.3.

6 BDM: the same continuity, a different polynomial space

Approximates: the same thing as RT — a flux. **Continuous:** the same thing — $\mathbf{v} \cdot \mathbf{n}$. **Lives in:** the same space — $H(\text{div})$. **Differs in:** which polynomials are kept, and therefore in the trade-off between accuracy in $\mathbf{v}$ and accuracy in $\text{div } \mathbf{v}$.

6.1 The element

The Brezzi–Douglas–Marini element takes the *complete* polynomial space of degree k ,

$$BDM_k(K) = (P_k(K))^2, \quad \dim BDM_k = (k+1)(k+2), \quad k \geq 1,$$

and imposes $H(\text{div})$ -conformity through the dofs rather than by restricting the polynomials. The normal trace of a $(P_k)^2$ field on an edge is a full polynomial of degree k in one variable, so it takes $k+1$ dofs per edge to pin it down:

$$\mathbf{v} \mapsto \int_F (\mathbf{v} \cdot \mathbf{n}) q \, ds, \quad q \in P_k(F).$$

For BDM_1 that is $3 \times 2 = 6$ dofs, exactly $\dim(P_1)^2$, so the lowest-order BDM element has no interior dofs at all.

6.2 The essential difference from RT

Both families are $H(\text{div})$ -conforming and both contain $(P_k)^2$. What separates them is the *divergence*:

$$\text{div } RT_k = P_k, \quad \text{div } BDM_k = P_{k-1}.$$

RT_k was constructed by adding just enough extra to $(P_k)^2$ to make the divergence surjective onto P_k ; BDM_k was not. The consequence for accuracy is a genuine trade-off:

Table 1. Lowest orders on a triangle. Both BDM_1 and RT_1 approximate the field itself to second order, but BDM_1 does it with six dofs instead of eight — at the price of one order in the divergence.

Space	dofs	contains	$\ \mathbf{v} - \mathbf{v}_h\ _{L^2}$	$\ \text{div}(\mathbf{v} - \mathbf{v}_h)\ _{L^2}$
RT_0	3	$(P_0)^2$	$O(h)$	$O(h)$
BDM_1	6	$(P_1)^2$	$O(h^2)$	$O(h)$
RT_1	8	$(P_1)^2$	$O(h^2)$	$O(h^2)$

6.3 When to prefer which

Choose RT_k when the divergence is part of the answer. If you need the mass balance resolved to the same order as the velocity — for instance when f is not piecewise constant and you want $\text{div } \mathbf{u}_h$ to track it accurately — the balanced approximation of RT is what you want. It pairs naturally with discontinuous P_k for the scalar unknown.

Choose BDM_k when you want the best field accuracy per degree of freedom. BDM_1 delivers the same $O(h^2)$ velocity as RT_1 from a smaller system. It pairs with discontinuous P_{k-1} .

In elasticity, where the unknown is a stress *tensor*, BDM rows are the standard building block: each row of the stress is an $H(\text{div})$ vector field, because the equilibrium equation $\text{div } \boldsymbol{\sigma} = -f$ applies the divergence row by row. The companion repository to this guide uses exactly this construction (Section 11.1).

There is also a structural way to say the difference, which Section 10 will make precise: BDM is the *full* polynomial family and RT is the *trimmed* one. They are the two natural members of the same construction, not rivals.

7 Nédélec elements: gluing the tangential component

Approximates: an electromagnetic field. **Dofs:** circulations along edges. **Continuous:** the tangential component. **Lives in:** $H(\text{curl})$. **Used for:** Maxwell's equations, eddy currents, waveguides, cavity resonances.

7.1 The problem that demands it

The time-harmonic Maxwell system, after eliminating the magnetic field, reads

$$\text{curl}(\mu^{-1} \text{curl} \mathbf{E}) - \omega^2 \varepsilon \mathbf{E} = \mathbf{J},$$

and its weak form is

$$\int_{\Omega} \mu^{-1} (\text{curl} \mathbf{E}) \cdot (\text{curl} \mathbf{F}) \, dx - \omega^2 \int_{\Omega} \varepsilon \mathbf{E} \cdot \mathbf{F} \, dx = \int_{\Omega} \mathbf{J} \cdot \mathbf{F} \, dx.$$

Every derivative enters through curl. By the rule of Section 2 the field belongs in $H(\text{curl})$, and the conforming discrete space must have continuous tangential components — nothing more.

7.2 The element

In two dimensions the lowest-order Nédélec element of the first kind is the 90° rotation of RT_0 :

$$N_0(K) = \{ \mathbf{v}(\mathbf{x}) = \mathbf{a} + b \mathbf{x}^\perp \}, \quad \mathbf{x}^\perp = (-x_2, x_1),$$

with the three dofs

$$\mathbf{v} \mapsto \int_{F_i} \mathbf{v} \cdot \mathbf{t}_i \, ds.$$

The same one-line miracle applies, rotated: along a straight edge the term $b \mathbf{x}^\perp$ has constant tangential component, so the tangential trace on each edge is a single shared number.

In three dimensions this becomes the *Whitney edge element*: one degree of freedom per mesh edge, equal to the circulation $\int_e \mathbf{E} \cdot \mathbf{t}$ of the field along that edge. On a tetrahedron that is six dofs. The name “edge element” comes from exactly this — the unknowns live on edges, not on vertices.

7.3 What tangential continuity means physically

Maxwell's interface conditions say it directly. Across a boundary between two materials:

- the *tangential* component of $\mathbf{E}$ is continuous (this is Faraday's law applied to a vanishingly thin loop straddling the interface);
- the *normal* component of $\mathbf{E}$ jumps, by the ratio of the permittivities, whenever surface charge is present.

An $H(\text{curl})$ -conforming element reproduces both statements exactly: it glues what physics glues and leaves free what physics leaves free.

7.4 Why Lagrange elements fail here — and they really do fail

This is the sharpest example in the guide of over-gluing causing actual damage. Discretising $\mathbf{E}$ with continuous vector Lagrange elements is not merely suboptimal; it produces wrong answers.

1. **It forbids a jump the physics requires.** At a dielectric interface the normal component must jump. A continuous vector space forces it to be continuous, so the discrete solution cannot approach the true one near any material interface.

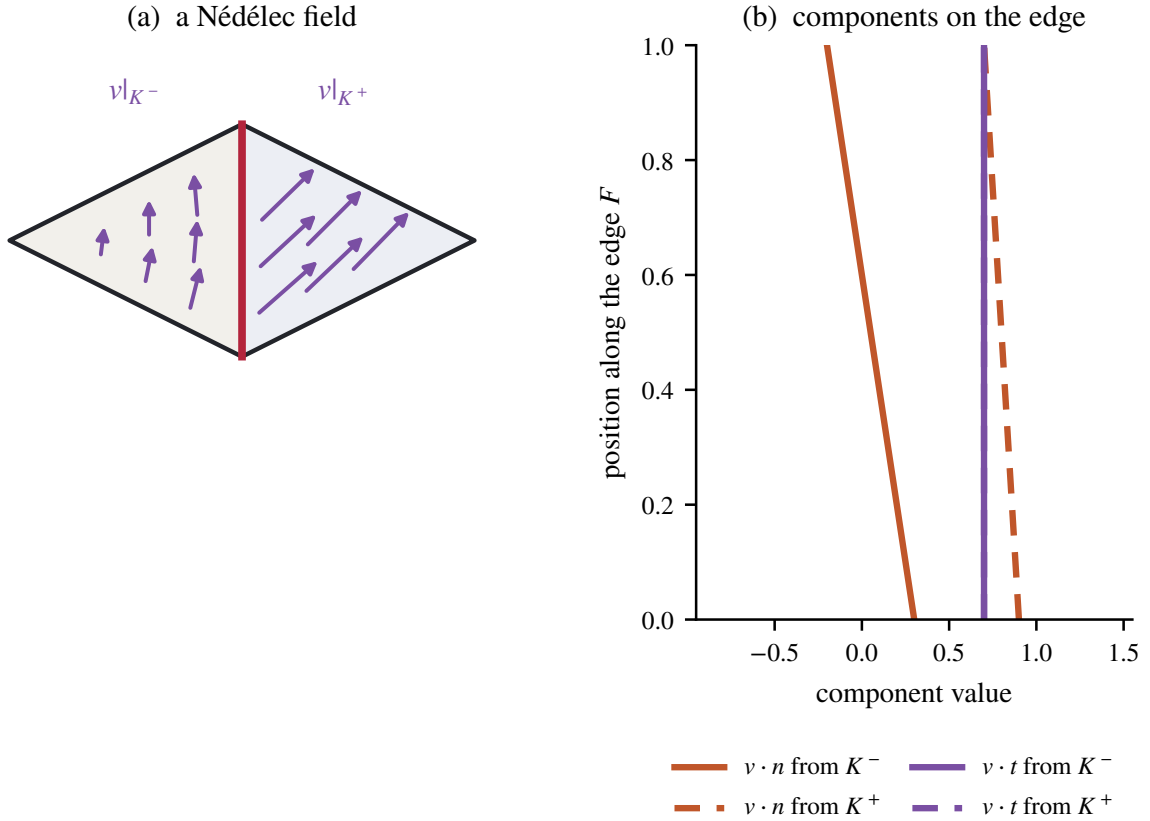


Figure 9. A Nédélec field on the same patch. Compare with Figure 7: now it is the two *tangential* curves (purple) that coincide, while the two *normal* curves (orange) are different functions. The roles of the components are exactly swapped.

2. **Spurious modes.** Computing cavity resonances with nodal elements produces extra, entirely non-physical eigenvalues scattered through the spectrum, which are hard to distinguish from real ones. The cause is structural: the operator curl has a huge kernel (every gradient), and a stable discretisation must reproduce the relation “curl-free = gradient” exactly at the discrete level. Nédélec spaces do, because $\nabla(P_k \text{ Lagrange}) \subset N_{k-1}$ exactly. Nodal vector spaces do not, and the mismatch surfaces as spurious eigenvalues.
3. **Re-entrant corners.** Near a re-entrant corner or an edge of a conductor the true field is singular: it lies in $H(\text{curl})$ but not in $(H^1)^d$. A nodal method searches for its solution inside $(H^1)^d$, which does not contain the answer, and converges confidently to the wrong function.

Point 2 is not a curiosity — it is the historical reason edge elements were invented, and it is the clearest illustration of the theme of Section 10: these element families are not an arbitrary collection, and using the wrong one breaks a structure you did not know you were relying on.

8 Four more elements worth knowing

8.1 Crouzeix–Raviart: continuity at one point per edge

The Crouzeix–Raviart element uses P_1 on each triangle, but places its three dofs at the *edge midpoints* rather than the vertices.

Two neighbouring cells now share one number per shared edge, not two. One number is not enough to determine a linear trace, so the traces from the two sides generally differ: the function is continuous at the midpoint of each edge and nowhere else along it. The global space is therefore *not* contained in H^1 .

Nonconforming methods

A method is *nonconforming* when $V_h \not\subset V$ — the discrete functions are not legal candidates for the continuous problem at all. This sounds fatal but is not, provided the violation shrinks fast enough under refinement. For Crouzeix–Raviart the saving grace is that, because the midpoint value of a linear function equals its average over the edge,

$$\int_F [[u_h]] \, ds = 0 \quad \text{on every edge } F,$$

so the jump has zero mean even though it is not zero. That is exactly enough for the consistency error to be $O(h)$ and the method to converge.

Why bother? Because this weaker gluing is precisely what some problems want:

- The pair (Crouzeix–Raviart velocity, piecewise constant pressure) is inf-sup stable for Stokes and remarkably cheap — an attractive combination that no low-order *conforming* pair achieves.
- In nearly incompressible elasticity, conforming low-order elements suffer *locking*: the incompressibility constraint uses up so many of the available degrees of freedom that the computed displacement is far too stiff. The extra freedom in the nonconforming space relieves this.

8.2 Hermite and Argyris: when H^1 is not enough

Fourth-order equations change the requirements entirely. For the biharmonic equation $\Delta^2 w = f$ — the model for the bending of a thin plate — the weak form pairs *second* derivatives:

$$\int_{\Omega} D^2 w : D^2 v \, dx = \int_{\Omega} f v \, dx.$$

Now the solution must live in H^2 , and running the jump calculation of Section 2 one order higher shows that a piecewise polynomial is in H^2 if and only if *both* the function and its normal derivative are continuous across every facet — that is, if and only if it is C^1 .

Building C^1 piecewise polynomials on triangles is genuinely hard, and it is easy to be misled:

The Hermite cubic triangle has 10 dofs: the value and both first derivatives at each of the three vertices, plus the value at the centroid. Despite carrying derivative dofs it is only C^0 : along a shared edge the two cells agree on the value and on the gradient *at the two endpoints*, but the normal derivative varies quadratically along the edge and the endpoint data do not determine it. Hermite triangles are therefore *not* H^2 -conforming — a common misconception worth stating plainly.

The Argyris triangle does the job. It uses quintic polynomials, with 21 dofs: at each vertex the value, both first derivatives and all three second derivatives ($6 \times 3 = 18$), plus the normal derivative at each edge midpoint (3). Since $\dim P_5 = 21$, this is unisolvent, and the resulting space really is C^1 and H^2 -conforming. It is expensive, and it is awkward on curved boundaries, but it is correct.

In practice many codes sidestep C^1 entirely — by rewriting the fourth-order problem as a mixed system of two second-order ones, or by using a C^0 interior penalty method that enforces normal-derivative continuity weakly, in the spirit of Section 4.

8.3 Taylor–Hood: elements come in pairs

The Stokes equations describe slow viscous flow:

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{f}, \quad \operatorname{div} \mathbf{u} = 0.$$

There are two unknowns, so there are two discrete spaces, and the new difficulty is that they must be chosen *together*. Here the pressure acts as a Lagrange multiplier enforcing $\operatorname{div} \mathbf{u} = 0$, and a multiplier space can be too rich for the space it constrains.

The inf-sup (LBB) condition

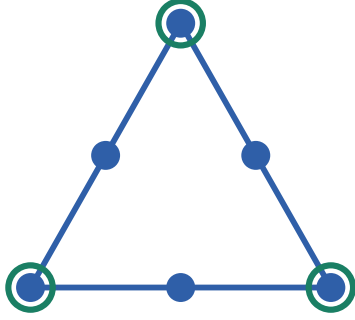
A mixed pair (V_h, Q_h) is stable if there is a constant $\beta > 0$, independent of h , with

$$\inf_{q_h \in Q_h} \sup_{\mathbf{v}_h \in V_h} \frac{\int_{\Omega} q_h \operatorname{div} \mathbf{v}_h \, dx}{\|\mathbf{v}_h\|_{H^1} \|q_h\|_{L^2}} \geq \beta.$$

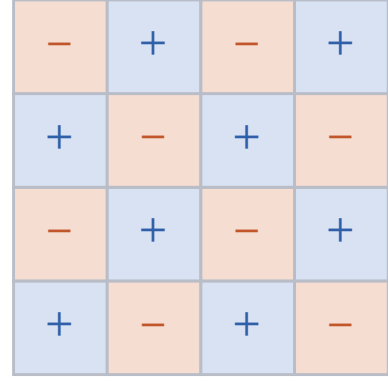
In words: for every possible pressure there must exist a velocity that “notices” it. If some nonzero q_h is invisible to every velocity in V_h , the discrete system has a spurious pressure mode and the computed pressure is garbage — even though the velocity may look fine.

The classical failure is equal-order P_1/P_1 , which admits a checkerboard pressure: a pattern alternating $+1$ and -1 from cell to cell that the discrete divergence simply cannot see. The Taylor–Hood pair, P_2 velocity with continuous P_1 pressure, is inf-sup stable, and is the standard workhorse for incompressible flow.

(a) the Taylor–Hood pair P_2/P_1



(b) a checkerboard pressure mode



an unstable pair such as P_1/P_1
cannot tell this apart from $p = 0$

- velocity: P_2 , all 6 nodes
- pressure: P_1 , vertices only

Figure 10. The Taylor–Hood pair: velocity dofs at all six P_2 nodes, pressure dofs at the three vertices only. Right: the checkerboard mode that destroys equal-order pairs. A stable pair must be rich enough in velocity to detect every pressure pattern.

This is the same requirement met earlier in disguise. RT_k with discontinuous P_k , and BDM_k with discontinuous P_{k-1} , are stable pairs for the mixed Poisson problem (2) for exactly this reason. Pairing spaces is not guesswork; it is what the next section is about.

9 One table for all of them

Table 2. The families at a glance. “Continuous quantity” is the single most useful column: it is what distinguishes the families, and everything else follows from it.

Element	Scalar / vector	Global space	Continuous quantity	Typical dofs	Typical PDE
Lagrange P_k	scalar	H^1	the whole value	point values at vertices, edge and interior nodes	Poisson, heat, elasticity
Crouzeix–Raviart	scalar	<i>none</i> (nonconforming)	value at edge midpoints only; jump has zero mean	value at each edge midpoint	Stokes, incompressible elasticity
Discontinuous P_k	either	L^2	nothing	cell-local nodal values	transport, conservation laws, mixed methods
RT_k	vector	$H(\text{div})$	normal component $\mathbf{v} \cdot \mathbf{n}$	normal moments on facets + interior moments	Darcy, mixed Poisson
BDM_k	vector	$H(\text{div})$	normal component $\mathbf{v} \cdot \mathbf{n}$	$k + 1$ normal moments per facet	mixed Poisson, stress in elasticity
Nédélec N_k	vector	$H(\text{curl})$	tangential component	circulations along edges	Maxwell, eddy currents, waveguides
Argyris P_5	scalar	H^2	value <i>and</i> normal derivative	value, gradient, Hessian at vertices; normal derivative at edge midpoints	biharmonic, plate bending
Taylor–Hood	pair	$H^1 \times H^1$	both values; stable by inf-sup	P_2 velocity nodes, P_1 pressure vertices	Stokes, Navier–Stokes

10 The de Rham complex: why this is a family, not a list

Everything so far could be read as a catalogue of unrelated tricks: someone needed normal continuity and invented RT ; someone else needed tangential continuity and invented Nédélec. That reading is wrong, and this section is the most important one in the guide.

10.1 The sequence

Consider the three classical differential operators of vector calculus in three dimensions, arranged in order:

$$H^1 \xrightarrow{\text{grad}} H(\text{curl}) \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2.$$

Each space is exactly the natural domain of the operator leaving it, and the image of each operator lands inside the domain of the next. Two identities you already know make the composition of consecutive arrows vanish:

$$\text{curl}(\text{grad } \phi) = 0, \quad \text{div}(\text{curl } \mathbf{v}) = 0.$$

More than that: on a domain without holes the converses hold too. A curl-free field *is* a gradient; a divergence-free field *is* a curl. In other words, at each stage the image of one arrow is *exactly* the kernel of the next. A sequence with that property is called *exact*, and this particular one is the *de Rham complex*.

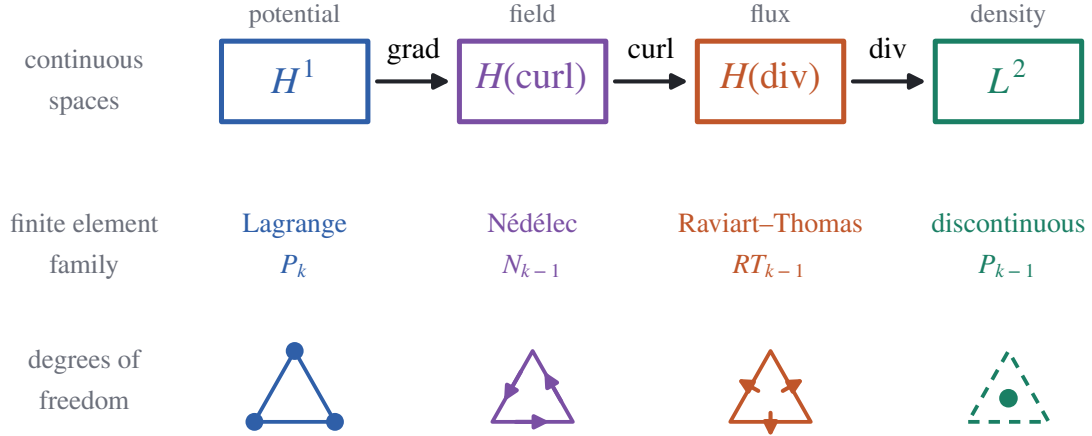
Read physically, it is a chain of familiar facts: a potential has a field; a field has a flux; a flux has a source density.

10.2 The discrete copy

Here is the point. The finite element families in this guide form a discrete sequence with the *same* structure:

$$\underbrace{P_k}_{\text{Lagrange}} \xrightarrow{\text{grad}} \underbrace{N_{k-1}}_{\text{Nédélec}} \xrightarrow{\text{curl}} \underbrace{RT_{k-1}}_{\text{Raviart–Thomas}} \xrightarrow{\text{div}} \underbrace{P_{k-1}}_{\text{discontinuous}} .$$

The gradient of a continuous piecewise polynomial really does land in the Nédélec space; the curl of a Nédélec function really does land in Raviart–Thomas; the divergence of a Raviart–Thomas function really does land in the discontinuous space — and the exactness property survives discretisation intact.



each map sends one space exactly onto the kernel of the next: $\text{curl grad} = 0$, $\text{div curl} = 0$

degrees of freedom sit on vertices $\rightarrow$ edges $\rightarrow$ faces $\rightarrow$ cells

Figure 11. The de Rham complex and its finite element realisation. Top: the continuous spaces and the operators linking them. Middle: the element family occupying each slot. Bottom: where the degrees of freedom sit — vertices, then edges, then faces, then cell interiors. The four families of this guide are the four slots of one sequence.

Notice the bottom row of Figure 11: the dofs march from vertices to edges to faces to cell interiors as you move along the sequence. That is not a coincidence either. In the language of differential forms, the four families are the finite element versions of 0-forms, 1-forms, 2-forms and 3-forms, and a p -form is naturally integrated over a p -dimensional piece of the mesh. Point values on vertices, circulations along edges, fluxes through faces, densities over cells.

10.3 What exactness buys

- **No spurious modes.** The Maxwell eigenvalue problem is well behaved precisely because the discrete kernel of curl equals the discrete gradients — which is discrete exactness at the second arrow. This is the structural reason behind Section 7, point 2.
- **Stable mixed methods.** The inf-sup condition of Section 8.3 for the pair $RT_k \times P_k$ follows from the divergence being surjective onto the discontinuous space — exactness at the third arrow. Choosing spaces from the same complex is a recipe for stability rather than a matter of luck.
- **Commuting interpolation.** Interpolating a function and then differentiating gives the same result as differentiating and then interpolating. Many error estimates fall out of this one diagram.

10.4 Two complexes in two dimensions, and where BDM sits

In two dimensions the sequence splits into two shorter ones, because curl has two different two-dimensional analogues:

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2, \quad H^1 \xrightarrow{\text{grad}} H(\text{curl}) \xrightarrow{\text{rot}} L^2,$$

where $\text{curl } \phi = (\partial_y \phi, -\partial_x \phi)$ is the rotated gradient and $\text{rot } \mathbf{v} = \partial_x v_2 - \partial_y v_1$ is the scalar curl. The second is the first with everything rotated by 90° — which is exactly the relationship between N_0 and RT_0 noted in Section 7.

Finally, BDM . The modern account, *finite element exterior calculus*, shows that each slot of the complex admits *two* natural polynomial families: a *trimmed* one and a *full* one. RT and Nédélec of the first kind are the trimmed family; BDM and Nédélec of the second kind are the full family. Table 1 was an instance of that general pattern, and it explains why the trade-off there looked so systematic.

This framework is due to Arnold, Falk and Winther. The same Richard Falk gives his name to the mixed elasticity element implemented in the companion repository — which is, as the next section shows, an application of exactly this way of thinking.

11 Putting it together

11.1 A worked example: one element, three spaces

The repository accompanying these notes solves linear elasticity in mixed form with weakly imposed symmetry, using the element introduced by Falk. It has three unknowns, and — the point of this guide in miniature — each one lives in a different tier of the hierarchy.

one mixed element, three different spaces

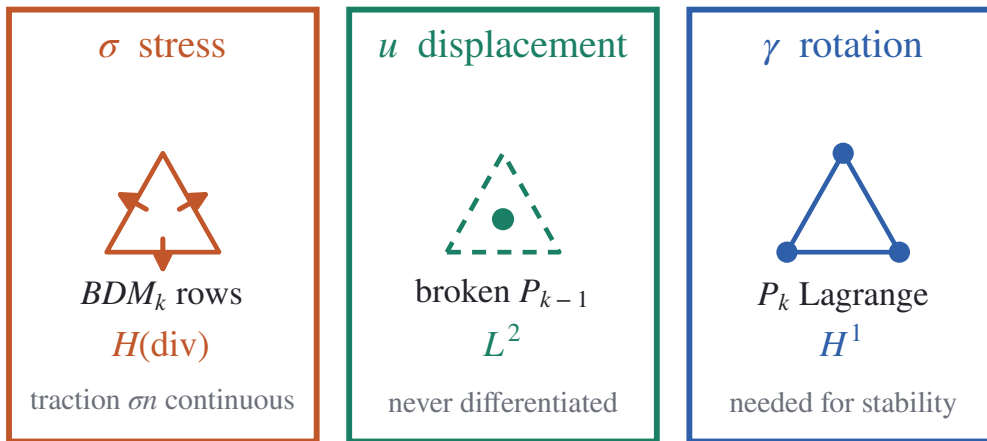


Figure 12. The three fields of the Falk mixed element, and the space each one needs. Reading the weak form tells you which is which.

The stress σ sits in $H(\text{div})$. The equilibrium equation $\text{div } \sigma = -f$ applies the divergence to each row of the stress tensor, so each row must be an $H(\text{div})$ vector field — in this implementation, a BDM_k field. The physical content of normal continuity here is Newton’s third law: the traction $\sigma \mathbf{n}$ that one side of an interface exerts on the other is equal and opposite, so it cannot jump.

The displacement u sits in L^2 . In the mixed form the displacement is never differentiated — the derivative has been moved onto the stress. By the logic of Section 4 there is no reason to ask for continuity, so u is a broken (discontinuous) P_{k-1} field.

The rotation γ sits in H^1 . This is the Lagrange multiplier enforcing the symmetry $\sigma = \sigma^\top$, and it is the subtle one: it is not differentiated either, yet it needs full H^1 continuity. The reason is stability, not regularity — the inf-sup argument for this element routes through a Stokes-like pairing that fails if the multiplier space is taken any larger.

That last point is a useful corrective to a rule stated too simply. “No derivative, no continuity” is the right first instinct, but stability can demand more, and in a mixed method the spaces must always be checked as a system rather than one at a time.

11.2 How to choose, in practice

- Write down the weak form first.** Not the PDE — the weak form. Integrate by parts the way you actually intend to, and see which derivative survives on which unknown.
- Match the space to the surviving derivative.** Full gradient $\Rightarrow H^1 \Rightarrow$ Lagrange. Only div $\Rightarrow H(\text{div}) \Rightarrow RT$ or BDM . Only curl $\Rightarrow H(\text{curl}) \Rightarrow$ Nédélec. No derivative $\Rightarrow L^2 \Rightarrow$ discontinuous.
- Ask what the true solution does at material interfaces.** If a component genuinely jumps there, any space that forbids the jump is disqualified regardless of what the derivative count said.
- For several unknowns, check the pair, not the pieces.** Take the spaces from one de Rham complex where you can; otherwise verify the inf-sup condition or use a documented stable pair.
- Then, and only then, pick a degree.** Higher k pays off when the solution is smooth; near corners and interfaces, mesh refinement is usually the better investment.

11.3 The one-page summary

	space	element	what is glued	typical PDE
	H^1	Lagrange P_k	the whole value	Poisson, elasticity, heat
	$H(\text{curl})$	Nédélec N_k	tangential part $v \cdot t$	Maxwell, eddy currents
	$H(\text{div})$	RT_k, BDM_k	normal part $v \cdot n$	Darcy, mixed Poisson
	L^2	discontinuous P_k	nothing	advection, shocks

reading down the table: less and less is glued together across element boundaries

Figure 13. The whole guide in one picture. Reading down: less and less is glued across element boundaries, the space gets larger, and the method asks less of the solution. The right element is the one that glues exactly what the equation — and the physics — says should be glued, and not one bit more.

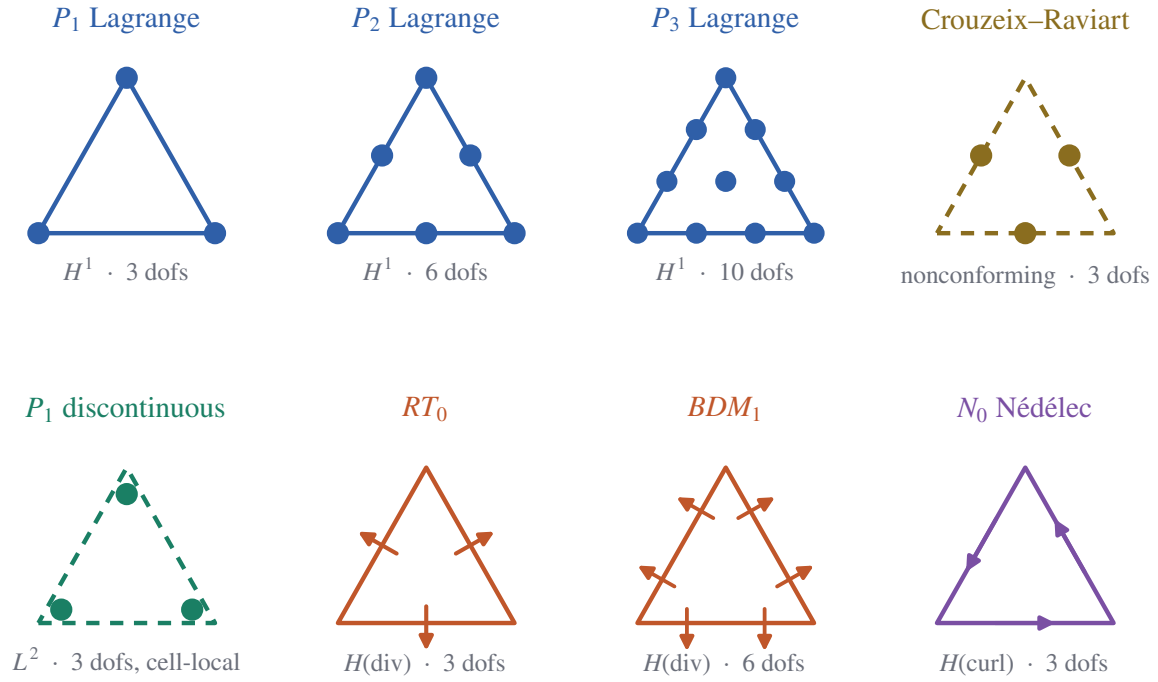


Figure 14. Degrees of freedom on the reference triangle. Filled circles are point values; arrows across an edge are normal moments; arrows along an edge are tangential moments; a dashed outline means nothing is shared with the neighbours. The position of the markers is the whole story: dofs on shared vertices and edges create continuity, dofs in the interior do not.

Where to go next

Run the demonstration. The script `examples/visualize_elements.py` in this directory draws the four behaviours of Figures 5, 6, 7 and 9 from explicitly constructed basis functions, with every space written out in a few lines of numpy. Reading it is a good way to convince yourself that nothing here is magic. See `README.md` for how to run it.

Standard references. Brenner & Scott, *The Mathematical Theory of Finite Element Methods*, for the general framework; Boffi, Brezzi & Fortin, *Mixed Finite Element Methods and Applications*, for everything mixed; Arnold, Falk & Winther, *Finite element exterior calculus* (Acta Numerica, 2006), for the structure of Section 10; Monk, *Finite Element Methods for Maxwell's Equations*, for edge elements.

In software. The DefElement project catalogues these elements with explicit basis functions and consistent conventions, and is the fastest way to check a definition or a dof count.